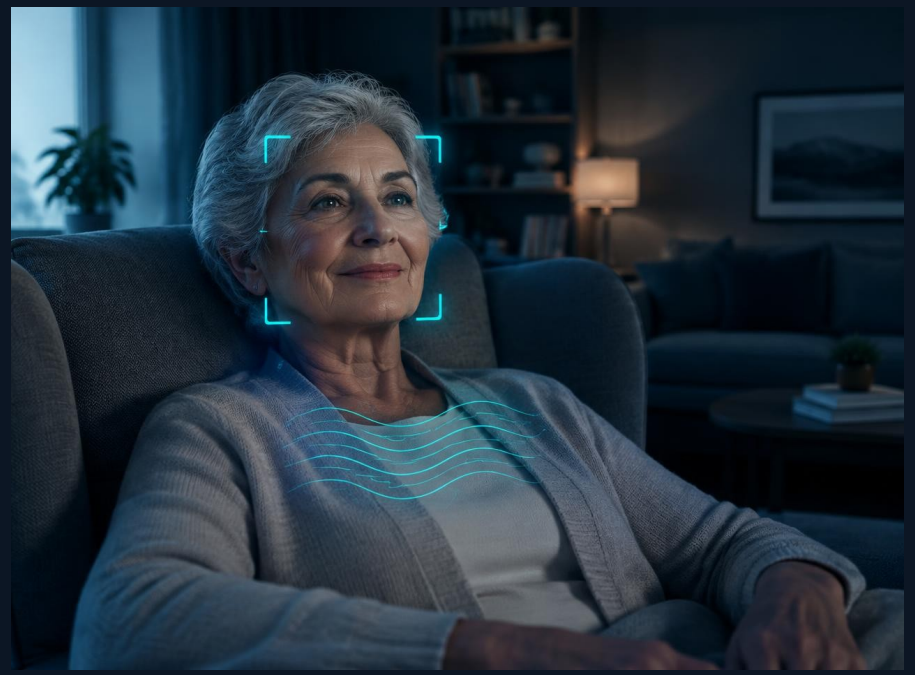
# RoomPulse AI

The Invisible Health Guardian · Vitals, No Wearable

## Contactless vital-sign intelligence from an ordinary camera.



Hossam Elshahaby

Browser-local processing. Transparent abstention.

# The problem: wearable dependency

Charging, fit, access, and adherence can interrupt monitoring.

01

## Independent living

Older adults may not consistently wear or charge sensors.

02

## Long periods at rest

People in beds or chairs need low-burden research tools.

03

## Trust matters

Missing or noisy signals must not become invented numbers.

# The solution: contactless intelligence

Estimate only when the observed signal supports a reading.

01 Track face and skin regions

02 Extract RGB and upper-body motion

03 Analyze POS, CHROM, or Green rPPG

04 Link every reading to its quality

GREEN Reliable signal conditions

AMBER Interpret with caution

RED No vital-sign number

Respiration remains experimental.

# Architecture and scientific validation

One analysis pipeline for live monitoring and dataset evaluation.

Camera → MediaPipe → skin RGB

Resample → detrend → POS / CHROM

Bandpass → Welch PSD → TrustScore

Local history → CSV / JSON / PDF

40 automated tests passed

Synthetic signals + integration checks

UBFC and CSV reference adapters

MAE, RMSE, bias, availability, Bland–Altman

No real-person accuracy or dataset benchmark results are claimed.

# Demonstration, impact, and next steps

A trustworthy sensing prototype—not a diagnostic promise.

Demo: camera tracking and honest rejection

Potential: home and bedside research

Privacy: no video upload or identity recognition

Next: synchronized reference validation

Broader skin-tone and lighting evaluation

Respiratory ground-truth studies

Motion robustness experiments

Thank you · Hossam Elshahaby